

BHARAT AI PLATFORM

Hyper-Local Fashion E-Commerce Marketplace & MSME Intelligence Engine Specification

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Target Audience: Technical Leads, Stakeholders & Auditors

Platform Scope: Customer, Seller & Admin Portals

1. Executive Summary & Core Architecture

Bharat AI is a next-generation hyper-local fashion e-commerce ecosystem specifically engineered for Indian consumers and MSME fashion sellers. The platform resolves traditional fashion discovery barriers by blending **voice & natural language AI**, **real-time weather/climate intelligence**, and **live demand-gap analytics** to empower small local sellers while delivering personalized shopping experiences to customers.

Layer	Technology Stack	Key Functional Role
Frontend UI	React.js (Vite), Tailwind CSS, Lucide Icons, Recharts	Responsive dark/light UI, interactive data charts, speech input
Backend API	Node.js, Express.js REST Framework	Authentication middleware, search orchestration, AI service integration
Database	PostgreSQL, Prisma ORM	Relational data management (Users, Products, Purchases, Reviews, Warnings)
AI Engine	Groq API (Llama 3.3 70B, Llama 3.1 8B, Mixtral)	Zero-shot product attribute auto-tagging, query intent parsing, multilingual summaries

2. Open-Source & Third-Party APIs / Libraries

The platform leverages high-performance open-source tools and free tier APIs to maximize reliability and speed:

API / Tool Name	Type / Ecosystem	Rationale & Business Purpose
Groq Llama 3 API	Open-Weights LLM Cloud Inference	Provides ultra-fast (<500ms) LLM inference for product auto-tagging and multi-lingual market analyst summaries across 6 Indian languages.
Web Speech API	Browser Native W3C Standard	Enables hands-free voice search in Indian regional accents directly in client browsers without audio recording server costs.
Prisma ORM	Open-Source DB Toolkit	Ensures type-safe SQL query generation, automated database migrations, and clean schema relationships.
Recharts	Open-Source React Visualization	Renders responsive sales trend lines, category distribution pie charts, and demand gap bar charts in Seller Hub.
Lucide React	Open-Source Icon Set	Delivers modern, lightweight SVG icons tailored for dark/light mode accessibility.
Weather & Climate Context	JSON Geo-Climate Service	Supplies real-time city temperature and climate indicators to dynamically filter weather-appropriate fashion shelves.

3. Customer Side Functionalities

A. Hyper-Local Personalized Homepage & Dynamic Shelves

- **Weather Shelf:** Automatically recommends clothing tailored to live city weather (e.g., lightweight cottons for summer, jackets for monsoon).
- **Local Boutiques Shelf:** Showcases items from verified local sellers within the user's city/state, marked with a 'Verified Local Seller' badge.
- **Budget-Friendly Shelf:** Segmented price shelves (Under ₹999, Under ₹1499, Under ₹1999) curated for cost-conscious shoppers.
- **Upcoming Festival Shelf:** Regional celebration recommendations mapped to active cultural events (e.g., Pongal, Diwali, Onam).

B. Search, Voice AI & Telemetry Engine

- **Multilingual Voice & Text Input:** Integrated mic button utilizing browser speech recognition for natural Indian language search.
- **AI Natural Language Intent Parsing:** Back-end LLM parses queries into category, gender, color, material, occasion, budget, and exclusions.
- **Smart Workplace Suitability Scoring:** Ranks office wear queries with a 0-100% Office Score while filtering out inappropriate items.
- **Internal Search Telemetry Debugger:** Displays real-time JSON intent extraction parameters directly on search results.

C. Order Execution, Reviews & UI Personalization

- **Mock Buy / Instant Checkout:** Complete checkout modal verifying pincode, calculating discounts, deducting stock, and persisting orders to DB.
- **Order History & Review System:** Customers can view past purchases, leave 1-5 star ratings, write text feedback, and flag complaints.
- **Theme & Location Picker:** Full Dark/Light mode context toggle and instant city location selector (e.g., Coimbatore, Tamil Nadu).

4. Seller Side Functionalities & Growth Hub

A. Product Catalog Management & AI Auto-Enrichment

- **AI-Powered Upload Form:** Sellers enter minimal basic fields (Name, Price, Image URL, Stock). The Llama 3 AI service automatically enriches product metadata with Material, Season, Climate Suitability, Ethnic Style, and Confidence score.
- **Products Management Table:** Full inventory control featuring status toggles (Active / Paused / Out of Stock), remaining stock counters, low-stock warnings, and inline price editing.

B. Market Search Gap Intelligence Cards (Core Feature)

The Market Intelligence Hub provides live competitive insights per seller region:

Metric / Feature	Implementation & Business Function
Live Searches Count	Aggregates real consumer search queries captured in the seller's city/state to measure raw demand volume.
Local Supply Count	Counts existing local products matching those consumer search terms.

Opportunity Score (0-100)	Algorithmically calculated ratio of search demand vs. available supply. Scores > 75 trigger high-demand alerts.
Demand Classification Badges	Visual indicators marking categories as 'HIGH DEMAND' (Green), 'BALANCED' (Blue), or 'SATURATED' (Amber).
Search Query Breakdown Modal	Interactive modal displaying exact search phrases typed by local buyers along with individual query counts.
Multilingual AI Analyst Summary	Groq Llama 3 powered business summary providing actionable stocking advice in 6 Indian languages (English, Hindi, Tamil, Telugu, Kannada, Bengali).

C. Sales Analytics & Category Requests

- **Revenue & Inventory KPI Cards:** 3 primary KPI metrics (Total Revenue ■, Units Sold, Total Orders) plus a Restock Alert notification banner.
- **Visual Trend Analytics:** Interactive line chart tracking revenue growth and pie chart illustrating sales distribution by fashion category.
- **Category Proposal Form:** Allows sellers to request unlisted fashion categories (with seasonal flags, handloom/traditional origin tags, and sample images) for admin approval.

5. Admin Platform & Governance Capabilities

The Admin Console ensures strict catalog moderation, quality control, and regional context management:

- **Platform Health Dashboard:** High-level KPI overview monitoring Total Platform Revenue, Registered Seller Count, Active Products, and Open Complaints.
- **Seller Moderation & Account Controls:** View seller profiles, track seller location metadata, and execute Account Block / Unblock actions.
- **Formal Seller Warning Issuance:** System for admins to issue formal policy warnings (stored in the SellerWarning model) regarding product issues or customer complaints.
- **Category Request Approval Workflow:** Admin review hub for seller category proposals, allowing one-click Approval or Rejection with feedback comments.
- **City Context & Climate Management:** Database controls (via CityContext) to configure city climate types, active regional festivals, and style preferences.

6. Platform Verification Summary

Feature Group	Status	Verification Notes
Customer Features	Verified (100%)	Weather shelf, Budget shelf, Festival shelf, Local Boutiques, Voice search, Checkout, Reviews, Dark/Light mode verified in codebase.
Seller Hub & AI Gap Engine	Verified (100%)	AI product upload, Gap Cards (Searches, Supply, Opportunity Score, Modal), 6-language summary, KPI cards, Category suggestions verified.
Admin Moderation	Verified (100%)	Seller block/unblock, formal warning notifications, category approvals, and city climate context controls verified.